SAMIUL SEIKH

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Portfolio | LeetCode | GeeksforGeeks | GitHub | LinkedIn

EDUCATION

B.Tech (Computer Science & Engineering)	2023-2027	Netaji Subhas University of Technology	8.05
CBSE (Class XII)	2023	School of Excellence, Rohini Sector 17	93.6%
CBSE (Class X)	2021	School of Excellence, Rohini Sector 17	88.0%

INTERNSHIP/S

Embedded & Systems Software Engineering Intern, Lumee AI Enterprises, Mumbai **May 2026 - July 2026**

- Built the embedded hardware-software stack for a real-time voice-and-image AI tutoring device (ESP32-S3: camera, I2S mic/speaker, SPI TFT, Wi-Fi) linked to a FastAPI/PostgreSQL backend over WebSockets.
- Structured firmware as **prioritized FreeRTOS tasks**, using **I2S DMA** for non-blocking audio alongside blocking SPI display writes, so real-time audio was not starved by display and network work.
- Cut voice-audio bandwidth ∼12× by switching both the mic uplink and speaker downlink from raw PCM to **Opus encoding**, keeping the voice usable on Wi-Fi links that raw PCM couldn't sustain.
- Replaced raw RGB565 frames with **grayscale JPEG** on both the display and camera-upload paths, shrinking frames ∼40-60× and removing bandwidth contention, so audio played **with no dropouts** while photos synced.
- Cut per-query time-to-first-audio on the backend from ∼**3 s to under 1 s** by loading data in parallel and streaming audio as it generated, and **eliminated under-load crashes** by bounding concurrent rendering and closing memory leaks.

TECHNICAL SKILLS

- **Languages & Tools:** C/C++, Python, Git, GitHub, Docker, VS Code
- **Embedded & RTOS:** ESP32, FreeRTOS, I2C, SPI, UART, I2S, DMA, Arduino
- **Systems & Networking:** Linux, Raspberry Pi, WebSockets, UDP, MQTT, Wi-Fi, REST APIs
- **Edge ML:** emlearn, TensorFlow Lite, ONNX, Scikit-learn

ACADEMIC PROJECTS

- **TI Smart Range Estimator: On-Device ML for EVs** | GitHub **June 2025 - July 2025**
  - Trained a **Random Forest** range predictor (scikit-learn) on a physics-based synthetic dataset with leakage-safe splitting and **5-fold cross-validation**.
  - Compiled the model to a <**100 KB** C header (emlearn) for <**5 ms** on-device inference on ESP32, with median fallbacks for sensor failures.
  - *Python (Scikit-learn), emlearn, C/C++, ESP32, Texas Instruments sensors*
- **TableMind: Offline AI Voice Assistant (ESP32 + Raspberry Pi 5)** | GitHub **March 2026**
  - Built a fully offline **5-thread voice pipeline** on a **Raspberry Pi 5** (openWakeWord → faster-whisper STT → Ollama LLM → Piper TTS), streamed to an ESP32 over **Wi-Fi UDP** with sub-5 ms IoT fast-path routing.
  - Engineered bidirectional audio on **ESP32** with **FreeRTOS** tasks and an NTP-synced clock.
  - *Python, C++, Raspberry Pi 5, ESP32, Ollama, faster-whisper, openWakeWord, Piper TTS, FreeRTOS*
- **IoT Home Automation System (Smart Lamp + RFID Smart Lock)** | GitHub **Oct 2024 - Present**
  - Built an end-to-end system of **ESP32-C3** devices and a **Flutter** app, with **mDNS** local discovery, REST APIs over Wi-Fi, and remote control via **MQTT (HiveMQ Cloud)**.
  - **Smart Lock:** unlocks via **RFID (MFRC522)** or over **Wi-Fi** from the app, with each entry **logged** in the app.
  - **Smart Lamp: 28 lighting patterns**, custom colors, and **MPU6050 motion-reactive** modes across 122 WS2812B LEDs, with OTA updates.
  - *C++ (ESP32), Dart (Flutter), MQTT, mDNS, REST APIs*

ACADEMIC ACHIEVEMENTS AND AWARDS

- **JEE Advanced, AIR 9,774:** ranked in the **top 1%** of candidates in India's most competitive engineering exam.
- **Finalist, NXP Autonomous Warehouse Challenge:** placed **top 14 nationally** among 5,800+ teams with an edge-AI navigation bot built on **ROS**, using **SLAM**.
- **Runner-up, FES NSUT AI FinTech Hackathon:** NLP-to-SQL tool that **cut dead-stock lookup time by 60%**.

POSITIONS OF RESPONSIBILITY

- **Director, ARES Robotics Society of NSUT** **2023 - Present**
  - Rose to **Director (2026)**; led technical ops and the **Moksha 2024** showcase.
  - Headed **Innovision 2025**, NSUT's annual technical fest, with **100+ registrations**, owning planning and execution end to end.

EXTRA-CURRICULAR ACTIVITIES AND ACHIEVEMENTS

- **NSS Volunteer, NSUT:** ran **7+ workshops (∼80 students each)** mentoring school students on engineering careers.